DDD

Data Definition Document

PulseAI

Data Definition Document

Complete schema reference for all 9 tables and collections

Document Type	Data Definition Document (DDD)
Version	1.0
Covers	5 SQL tables + 4 MongoDB collections
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1 Document Overview

This Data Definition Document (DDD) is the single source of truth for all data in PulseAI. It describes every table, collection, field, data type, valid values, and the business meaning of each piece of data.

It covers 5 SQL tables stored in Supabase (PostgreSQL) and 4 document collections stored in MongoDB Atlas — a total of 9 data sources, all linked by the customer_id field.

2 How to Read This Document

Each section covers one table or collection. For SQL tables, you will find a field-by-field breakdown with: the data type, whether the field can be empty (Nullable), whether it is a Primary Key (PK) or Foreign Key (FK), a plain-English description of what the field means, and a real example value.

For MongoDB collections, fields use dot notation to show nesting. For example, events[].event_type means the event_type field inside each item in the events array.

Symbol	Meaning
PK	Primary Key — uniquely identifies each row
FK	Foreign Key — links to another table's PK
—	Regular data field (not a key)
Yes (Nullable)	Field can be empty / null
No (Nullable)	Always required — never null

3 Data Architecture Summary

Database	Type	Tables / Collections	Records	Purpose
Supabase	PostgreSQL	customers, transactions, subscriptions, risk_scores, model_training_labels	~4,100	Structured: profiles, payments
MongoDB Atlas	MongoDB	customer_events, support_tickets, behavioral_sessions, notification_responses	~3,172 docs	Unstructured: events, chats, s

SQL Tables — Supabase (PostgreSQL)

4 customers

customers

Supabase PostgreSQL | 100 records (synthetic)

The master record for every customer. Every other table and collection links back here via customer_id. This tells you who the customer is, how they joined, and what plan they are on. Think of it as the customer profile card.

Field Name	Data Type	Nullable	Key	Description	Example
customer_id	VARCHAR	No	PK	Unique identifier. Format: CUST-XXXXXXXX. The master key that links every table and MongoDB collection.	*CUST-B1A7E63F*
full_name	VARCHAR	No	—	Customer full name (first + last). Pseudonymised in production.	*Katherine Anderson*
age	INT	No	—	Customer age in years at signup. Range: 18–80 in this dataset.	*23*
gender	VARCHAR	Yes	—	Self-reported gender. Values: Male, Female, Non-binary.	*Female*
location_city	VARCHAR	Yes	—	City where the customer is based. Used for geographic analytics.	*Indianapolis*
location_country	VARCHAR	Yes	—	Country name. Used for regional segmentation.	*US*
acquisition_channel	VARCHAR	No	—	How the customer found the product. Values: organic_search, paid_ad, referral, social_media, email_campaign, app_store, direct, partner.	*social_media*
plan_type	VARCHAR	No	—	Current subscription plan. Values: basic ($29.99), standard ($59.99), premium ($99.99), enterprise ($249.99).	*basic*
tenure_days	INT	No	—	Days since account creation as of the data export date. Computed from created_at.	*2312*
created_at	TIMESTAMP	No	—	Date and time the account was created (UTC). The start of the customer journey.	*2020-02-24 00:00:00*
is_active	BOOLEAN	No	—	Whether the customer has an active subscription. TRUE = active, FALSE = churned.	*True*

Table — customers field definitions

5 transactions

transactions

Supabase PostgreSQL | 3,847 records (synthetic)

Every billing event — one row per monthly payment cycle per customer. This is where you see whether a customer pays on time, fails payments, or goes pending. Failed payment patterns in the last 3 months are one of the strongest churn predictors the model uses.

Field Name	Data Type	Nullable	Key	Description	Example
transaction_id	VARCHAR	No	PK	Unique payment event identifier. Format: TXN-XXXXXXXXXX.	*TXN-70B62993-A*
customer_id	VARCHAR	No	FK	Links to customers.customer_id. Identifies which customer made this payment.	*CUST-B1A7E63F*
amount_due	DECIMAL	No	—	Amount the customer was billed this cycle (USD). Varies by plan. Occasional discounts (10-20%) reduce this.	*29.99*
amount_paid	DECIMAL	No	—	Amount actually received. Equals amount_due on success. 0.00 on failed or pending.	*29.99*
payment_status	VARCHAR	No	—	Payment outcome. success = received. failed = declined or errored. pending = awaiting processing. Key ML feature.	*success*
payment_method	VARCHAR	No	—	How customer paid. Values: credit_card, debit_card, paypal, bank_transfer, invoice.	*credit_card*
due_date	TIMESTAMP	No	—	Date the payment was due — first day of the billing cycle.	*2020-02-24 00:00:00*
paid_at	TIMESTAMP	Yes	—	When payment was received. NULL if failed or pending. Typically 0-3 days after due_date.	*2020-02-24 14:46:24*

Table — transactions field definitions

6 subscriptions

subscriptions

Supabase PostgreSQL | 150 records (synthetic)

The history of what plan each customer has been on. A customer can have multiple rows — one per subscription period. If they started on basic, upgraded to standard, then cancelled, they would have three rows. A downgraded status is a strong early warning signal of coming churn.

Field Name	Data Type	Nullable	Key	Description	Example
subscription_id	VARCHAR	No	PK	Unique identifier for this subscription period. Format: SUB-XXXXXXXXXX.	*SUB-EA42BCF2-F*
customer_id	VARCHAR	No	FK	Links to customers.customer_id.	*CUST-B1A7E63F*
plan_name	VARCHAR	No	—	The plan during this period. Values: basic, standard, premium, enterprise.	*basic*
plan_start_date	DATE	No	—	First day on this plan.	*2020-02-24*
plan_end_date	DATE	No	—	Last day on this plan — either plan change date, cancellation date, or next billing date for active customers.	*2023-11-19*
renewal_status	VARCHAR	No	—	What happened at end of period. Values: auto, manual, upgraded, downgraded, cancelled. downgraded and cancelled are churn risk signals.	*upgraded*
last_upgraded_at	TIMESTAMP	Yes	—	Timestamp of most recent upgrade. NULL if never upgraded.	*2023-11-19*
last_downgraded_at	TIMESTAMP	Yes	—	Timestamp of most recent downgrade. NULL if never downgraded. A non-null value is a strong churn risk signal.	—

Table — subscriptions field definitions

7 risk_scores

risk_scores

Supabase PostgreSQL | Populated after model training

This table is written by PulseAI — it is the output table. Every time the inference engine scores a customer, a row is inserted here. The business reads PulseAI results by querying this table rather than calling the model directly.

Field Name	Data Type	Nullable	Key	Description	Example
score_id	VARCHAR	No	PK	Unique identifier for this scoring event. Auto-generated UUID.	*SCR-A1B2C3D4*
customer_id	VARCHAR	No	FK	Links to customers.customer_id.	*CUST-B1A7E63F*
risk_score	FLOAT	No	—	Model output probability of a negative event. Range: 0.0 (safe) to 1.0 (certain churn). The sigmoid output of the PyTorch classifier.	*0.87*
risk_label	VARCHAR	No	—	Human-readable label from risk_score. Thresholds: low (0.0-0.3), medium (0.3-0.6), high (0.6-0.8), critical (0.8-1.0).	*high*
predicted_event	VARCHAR	No	—	Type of event predicted. Values: churn, default, lapse, abandon.	*churn*
model_version	VARCHAR	No	—	Version of the model that scored this customer. Format: semver e.g. 1.0.0. Enables model comparison over time.	*1.0.0*
scored_at	TIMESTAMP	No	—	When this score was computed (UTC). Used to find the most recent score per customer.	*2026-06-24 10:30:00*

Table — risk_scores field definitions

8 model_training_labels

model_training_labels

Supabase PostgreSQL | Populated during training setup

The ground truth table — used only when training the PyTorch model. It records what actually happened to each historical customer. The model learns by comparing its predictions against these labels. This table is NOT used during inference.

Field Name	Data Type	Nullable	Key	Description	Example
label_id	VARCHAR	No	PK	Unique identifier for this training label record.	*LBL-001*
customer_id	VARCHAR	No	FK	Links to customers.customer_id.	*CUST-B1A7E63F*
event_type	VARCHAR	No	—	The type of event being predicted. Values: churned, defaulted, lapsed.	*churned*
event_occurred	BOOLEAN	No	—	Did the event happen? TRUE = yes (positive label for training). FALSE = no (negative label).	*True*
event_date	DATE	Yes	—	Date the event occurred. NULL if event_occurred is FALSE.	*2024-03-15*
observation_end_date	DATE	No	—	Features are built from data before this date; label is what happened after. Prevents data leakage.	*2024-02-15*

Table — model_training_labels field definitions

MongoDB Collections — MongoDB Atlas

9 customer_events

customer_events

MongoDB Atlas | 100 documents, 4,768 total events

The most important collection for the ML model. Each document contains a chronological array of every action a customer has taken. The sequence of events is what the PyTorch LSTM reads to detect risk patterns — login, complaint, cancelled page visit, inactivity, and so on.

Field / Path	Type	Description	Example
_id	String	Unique document ID for MongoDB. Format: EVTDOC-XXXXXXXXXXXX.	*EVTDOC-A1B2C3*
customer_id	String	Links to customers table in Supabase. The join key.	*CUST-B1A7E63F*
risk_profile	String	Pre-computed risk level. Values: low, medium, high.	*low*
events	Array	Chronological array of all customer actions. 10-80 events per customer.	*[{...},{...}]*
events[].event_id	String	Unique ID for this specific event. Format: EVT-XXXXXXXXXXXX.	*EVT-001*
events[].event_type	String	What the customer did. E.g.: login, payment_failed, cancel_page_visited, email_ignored, inactivity. Encoded as integers for the LSTM model.	*payment_failed*
events[].timestamp	String	ISO-8601 datetime when event occurred (UTC). Sorted chronologically — order is critical for the LSTM.	*2024-01-05T14:30:00Z*
events[].channel	String	How customer was interacting. Values: mobile_app, web, tablet_app.	*mobile_app*
events[].metadata	Object	Flexible key-value data specific to this event type. E.g. a login has device and os. A payment_failed has amount and reason. Structure varies — this is why MongoDB is used.	*{reason: insufficient_funds}*

Collection — customer_events field definitions

10 support_tickets

support_tickets

MongoDB Atlas | 317 documents

Every customer support interaction. Each document is one ticket with the full conversation thread as a messages array. High-risk customers generate more tickets, more billing complaints, and more escalations. Sentiment score summarises the emotional tone of the conversation.

Field / Path	Type	Description	Example
_id	String	Unique document ID. Format: TICKET-XXXXXXXXXXXX.	*TICKET-001*
customer_id	String	Links to customers table. Join key.	*CUST-B1A7E63F*
category	String	Type of issue. Values: billing, technical, account_access, feature_request. Billing is most common for high-risk customers.	*billing*
priority	String	Urgency level. Values: low, medium, high.	*high*
status	String	Current state. Values: open, resolved, closed. More open tickets = higher risk.	*resolved*
escalated	Boolean	Whether escalated to senior agent. TRUE is a strong churn signal — 35% of high-risk customers have escalations.	*true*
created_at	String	ISO-8601 datetime ticket was raised (UTC).	*2024-01-03T10:00:00Z*
resolved_at	String	ISO-8601 datetime ticket was resolved. NULL if still open.	*2024-01-04T15:00:00Z*
sentiment_score	Float	Automated sentiment: -1.0 (very negative) to +1.0 (very positive). High-risk customers score -0.9 to -0.3.	*-0.76*
messages	Array	Full conversation thread. Each item is one message.	*[{...}]*
messages[].sender	String	Message author. Values: customer, agent.	*customer*
messages[].text	String	Message content. Used for NLP and sentiment analysis.	*I was charged twice*
messages[].timestamp	String	ISO-8601 datetime of this message.	*2024-01-03T10:00:00Z*
agent_id	String	ID of the agent who handled the ticket. Format: AGT-XXXXXXXXXXXX.	*AGT-X9Y8Z7*
resolution_note	String	Closing note. NULL if still open.	*Issue resolved.*

Collection — support_tickets field definitions

11 behavioral_sessions

behavioral_sessions

MongoDB Atlas | 2,655 documents

Every browsing session on web or mobile app. Each document records which pages were visited, time spent, and actions taken. The key risk signal is the cancel_plan page — customers spending significant time there are strong churn candidates.

Field / Path	Type	Description	Example
_id	String	Unique session document ID. Format: SES-XXXXXXXXXXXX.	*SES-001*
customer_id	String	Links to customers table. Join key.	*CUST-B1A7E63F*
session_start	String	ISO-8601 datetime session began (UTC).	*2024-01-08T19:00:00Z*
session_end	String	ISO-8601 datetime session ended (UTC).	*2024-01-08T19:14:00Z*
duration_seconds	Integer	Total session length in seconds.	*840*
channel	String	Access method. Values: mobile_app, web, tablet_app.	*mobile_app*
device	String	Device used. E.g.: iPhone 14, Samsung Galaxy S23, MacBook Pro.	*iPhone 14*
os	String	Operating system. Values: iOS 17, Android 14, macOS Sonoma, Windows 11.	*iOS 17*
pages_visited	Array	Ordered list of pages visited in this session.	*[{...}]*
pages_visited[].page	String	Page name. Key risk pages: cancel_plan, payment_retry, downgrade_plan. Others: home, billing, dashboard, settings, support.	*cancel_plan*
pages_visited[].entered_at	String	Datetime customer landed on this page.	*2024-01-08T19:07:00Z*
pages_visited[].time_spent_sec	Integer	Seconds on this page. Long time on cancel_plan (>60s) is a strong risk signal.	*390*
pages_visited[].actions_taken	Array	Actions on this page. E.g. on cancel_plan: viewed_retention_offer, clicked_cancel, exited_without_cancel.	*[viewed_cancel_page]*
total_pages	Integer	Total pages visited in session.	*4*
bounced	Boolean	TRUE if only 1 page visited and under 15 seconds — very low engagement.	*false*
risk_signals	Object	Pre-computed flags summarising risk content of this session for fast feature access.	*{...}*
risk_signals.visited_cancel_page	Boolean	TRUE if cancel_plan was visited.	*true*

Field / Path	Type	Description	Example
risk_signals.visited_billing_page	Boolean	TRUE if billing was visited.	*false*
risk_signals.visited_downgrade_page	Boolean	TRUE if downgrade_plan was visited.	*false*
risk_signals.payment_retry_attempt	Boolean	TRUE if payment_retry was visited.	*false*

Collection — behavioral_sessions field definitions

12 notification_responses

notification_responses

MongoDB Atlas | 100 documents, 739 total notifications

All marketing and system notifications sent to customers and whether they responded. One document per customer. Declining engagement — dropping open rates, emails ignored, unsubscribes — is a classic early-warning signal before churn.

Field / Path	Type	Description	Example
_id	String	Unique document ID. Format: NRDOC-XXXXXXXXXXXX.	*NRDOC-001*
customer_id	String	Links to customers table. Join key.	*CUST-B1A7E63F*
engagement_summary	Object	Pre-computed engagement metrics — fast-access features for the ML model.	*{...}*
engagement_summary.total_sent	Integer	Total notifications sent across all channels.	*12*
engagement_summary.total_opened	Integer	How many were opened or tapped.	*8*
engagement_summary.total_clicked	Integer	How many resulted in a click-through action.	*3*
engagement_summary.open_rate	Float	total_opened / total_sent. High-risk avg: 0.25-0.40. Low-risk avg: 0.75-0.88.	*0.667*
engagement_summary.click_rate	Float	total_clicked / total_sent. Measures how actionable notifications were.	*0.250*
engagement_summary.unsubscribed	Boolean	TRUE if customer unsubscribed from any email. Very strong churn predictor.	*false*
notifications	Array	All individual notification records for this customer.	*[{...}]*
notifications[].notification_id	String	Unique notification ID. Format: NOTIF-XXXXXXXXXXXX.	*NOTIF-001*
notifications[].type	String	Channel. Values: email, sms, push, in_app.	*email*
notifications[].sent_at	String	ISO-8601 datetime sent (UTC).	*2024-01-01T08:00:00Z*
notifications[].subject	String	Email subject line. NULL for sms/push/in_app.	*Your bill is due*
notifications[].message	String	Text for sms/push/in_app. NULL for email.	*Payment due in 3 days*
notifications[].opened	Boolean	Whether the notification was opened/read.	*false*
notifications[].clicked	Boolean	Whether the customer clicked through.	*false*
notifications[].converted	Boolean	Whether the customer completed the desired action after clicking.	*false*

Field / Path	Type	Description	Example
notifications[].unsubscribed	Boolean	For email only: TRUE if customer unsubscribed via this specific email.	*false*

Collection — notification_responses field definitions

13 Data Relationships & Join Keys

All 9 data sources are linked by a single field: customer_id. This is the primary key in the customers table and appears as a reference in every other table and collection. When the feature pipeline builds the ML dataset, it joins all sources on this field to create one complete feature vector per customer.

Source A	Source B	Join Field	Relationship
customers	transactions	customer_id	One-to-Many (1 customer, many transactions)
customers	subscriptions	customer_id	One-to-Many (1 customer, 1+ plan periods)
customers	risk_scores	customer_id	One-to-Many (1 customer, many score events)
customers	model_training_labels	customer_id	One-to-One (1 customer, 1 label)
customers	customer_events	customer_id	One-to-One (1 customer, 1 events doc)
customers	support_tickets	customer_id	One-to-Many (1 customer, many tickets)
customers	behavioral_sessions	customer_id	One-to-Many (1 customer, many sessions)
customers	notification_responses	customer_id	One-to-One (1 customer, 1 notifs doc)

Table — cross-source join relationships

14 Value Enumerations Reference

The following fields have a fixed set of allowed values. Any value outside these lists should be treated as a data quality error and logged for investigation.

Field	Source	Allowed Values
plan_type	customers	basic, standard, premium, enterprise
acquisition_channel	customers	organic_search, paid_ad, referral, social_media, email_campaign, app_store,
payment_status	transactions	success, failed, pending
payment_method	transactions	credit_card, debit_card, paypal, bank_transfer, invoice
renewal_status	subscriptions	auto, manual, upgraded, downgraded, cancelled
risk_label	risk_scores	low, medium, high, critical
predicted_event	risk_scores	churn, default, lapse, abandon
event_type	customer_events	login, login_failed, page_view, feature_used, payment_success, payment_fail
category	support_tickets	billing, technical, account_access, feature_request
priority	support_tickets	low, medium, high
status	support_tickets	open, resolved, closed
page	behavioral_sessions	home, dashboard, billing, cancel_plan, plan_comparison, settings, support, pa
type	notification_responses	email, sms, push, in_app

Table — enumeration reference for all constrained fields

End of Document — PulseAI Data Definition Document v1.0